# From the Lab to the Real World:
# Building Usable Socially Assistive Robots


## SOOMIN SHIN

Personal Website ⋄ Google Scholar ⋄ Github

Email: soomin.shin@uwaterloo.ca


Ph.D. candidate specializing in **Human-Robot Interaction (HRI) and LLM-driven Social Robots**, focused on building readily deployable systems for real-world settings that can be used by non-technical users. Expert at designing, prototyping, and deploying socially assistive robots with advanced interaction capabilities. Passionate about translating research into accessible, field-ready systems, with a strong publication record in top-tier venues and hands-on experience in collaborative, interdisciplinary environments.

## EDUCATION

**University of Waterloo** - *PhD. in Electrical and Computer Engineering* *Present - 09/2027 Expected*
*Supervisor: Prof. Kerstin Dautenhahn*

**Korea University** - *MSc. in Brain and Cognitive Engineering*

**Seoul Women's University** - *B.E. in Multimedia, B.A. in Child Studies*

## RESEARCH EXPERIENCE

**Graduate Researcher** Social and Intelligent Robotics Research Lab, University of Waterloo

- **Research Focus**: Developing social robots for the real clinical setting [1], [2]
- Designed and implemented a system for a **general-purpose therapeutic robot**, engineering an architecture that integrates LLMs with a robotic platform using PyTorch and ROS.
- Generated expressive, goal-oriented robot gestures using the MoveIt Motion Planning Framework and inverse kinematics, improving the robot's non-verbal communication.
- Developed a scalable, deployable prototype by translating ambiguous user needs into a concrete system design, leading a two-year development cycle in collaboration with a team of clinical experts.
- Developed a **physical robotic system within a clinical setting**, managing the full-stack development of the robot software and a Flask-based web interface for therapists.
- **Applied data-driven insights from user studies**: to validate system performance and iteratively improve algorithmic efficiency and operational effectiveness.

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**Research Intern** Korea Institute of Science and Technology

- **Heterogeneous Robot Services**: Developed and evaluated a robot-assisted system for isolation wards, incorporating telemedicine, emergency alerts, and delivery robots [3].
- **Trust in Robot Hierarchy**: Analyzed how perceived hierarchy in a robot team impacts user trust and service evaluations [4].
- **User-Control vs. Autonomy**: Studied users' preference for a reactive robot that follows verbal commands over a proactive system [5].

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**Graduate Researcher** Cognitive Systems Lab, Korea University

- **Research Focus**: Explored how contextual information influences human and neural network model on emotion recognition. Found contextual information significantly alters emotion perception in humans and AI systems. [6].
- **Human-Model Comparison Study**: Designed experiments to compare discrepancies between human perception and model predictions on contextual emotion recognition.
- **Contextual Emotion Model Evaluation**: Assessed the performance of a pre-trained CNN-based model on the collected In-the-Wild Dataset to analyze its effectiveness in recognizing contextual emotions.

## PUBLICATIONS


[1] S. Shin, S. Chandra, A. Rajan, S. Shah, and K. Dautenhahn, "How co-design and personas can inform game implementation for robot-assisted speech therapy in clinical settings," *Submitted to IEEE International Conference on Robot and Human Interactive Communication (RO-MAN)*, 2025.

[2] N. Azizi, S. Chandra, A. Rajan, *et al.*, "Development of robot-assisted speech-language therapy: Co-design with speech-language pathologists," in *Proceedings of the 16th International Conference on Social Robotics (ICSR 2024)*, Springer, 2024.

[3] Y. Kwon, S. Shin, K. Yang, *et al.*, "Heterogeneous robot-assisted services in isolation wards: A system development and usability study," in *2023 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, IEEE, 2023, pp. 8069–8076.

[4] S. Shin and S. S. Kwak, "Do hierarchies in a robot team impact the service evaluation by users?" In *2023 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, IEEE, 2023, pp. 3983–3990.

[5] S. Shin, D. Kang, and S. S. Kwak, "Is a robot trustworthy enough to delegate your control?" In *2023 32nd IEEE International Conference on Robot and Human Interactive Communication (RO-MAN)*, IEEE, 2023, pp. 2415–2420.

[6] S. Shin, D. Kim, and C. Wallraven, "Contextual modulation of affect: Comparing humans and deep neural networks," in *Companion Publication of the 2022 International Conference on Multimodal Interaction*, 2022, pp. 127–133.


## TECHNICAL SKILLS

**Programming Languages:** Python (Proficient), JavaScript (Proficient), C/C++ (intermediate)

**Robotics and AI:**

Frameworks: Robot Operating System (ROS1), NVIDIA Riva, PyTorch

Robots: QT robot, Vector

## ACHIEVEMENTS

**Graduate Research Studentship** — *Present*
*University of Waterloo*

**International Doctoral Student Award** — *Sep 2023*
*University of Waterloo*

**Provost's Doctoral Entrance Award for Women** — *Sep 2023*
*University of Waterloo*

## TEACHING & MENTORING EXPERIENCE

**Robotics Programming Mentoring** — *University of Waterloo, Fall 2024 - Present*

**Teaching Assistant, University of Waterloo**

Fundamentals of Computational Intelligence (ECE457B) (Spring 2025)

Programming for Performance (ECE459) (Winter 2025)

Digital Computation (BME121) (Fall 2024)

Algorithm Design and Analysis (ECE406) (Winter 2024)